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NIHR Invention for Innovation Programme

Scheme	i4i THRIVE
Funding Opportunity	2025/288 i4i Translate Healthcare Research through InnoVation and Entrepreneurship (THRIVE) Researcher-led

SUMMARY INFORMATION

Application Title	Neonatal Wireless Transmission System: Commercial Product to Translation into Clincal Practice
Contracting Organisation	CAMBRIDGE UNIVERSITY HOSPITALS NHS FOUNDATION TRUST
Contracting Organisation Department	Paediatrics
Start Date	01/11/2025
End Date	31/07/2026
Duration (months)	9
Total Cost to NIHR	£140,962.84
NHS Support Costs	£0.00
NHS Excess Treatment Costs	£0.00
Non-NHS Excess Treatment Costs	£0.00
Has this application been previously submitted to this or any other funding body?	No

LEAD APPLICANT & RESEARCH TEAM

Lead Applicant

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Full Name	Dr Kathryn Beardsall
Organisation	University of Cambridge
Department	Paediatrics
Proposed Role	I am a Consultant Neonatologist, with experience in leading neonatal studies, ranging from small feasibility studies to large off label device and drug trials. I am clinical lead for the development of the NEWTS system. I am providoing clinical expertise to support the development of critical aspects of the utility, efficacy, and safety of the system, as well as design and execution of the clinical study. I am well-placed to engage with the multidisciplinary clinical teams and the wider neonatal community both in the UK and abroad, ensuring the device is optimized for the clinical setting and the future route to market. However, the route to commercialization is hugely challenging, especially in paediatrics and intensive care. In this application I wish to have time to gain personal knowledge and training in the translational route for implementation of the NEWTS system into a commercially viable business. This will require gaining experience and understanding of different commercial strategies and opporuntunities to engage with key policy stakeholders which the THRIVE programm offers. This will support the translation of the NEWTS system into clinical care but also helps to build a team developing medical technology. As a co-theme lead for the mind and body theme of the NIHR HealthTech Research Centre in Paediatrics and Child Health I am keen use this experience to support other collaborations and and colleagues so that children can benefit in the future from innovations in care.
ORCID	0000-0003-3582-183X
% FTE Commitment	20.00%

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Co-Applicants

Full Name	Dr Sarah Lindsay
Organisation	University of Cambridge
Department	Department of Paediatrics
Proposed Role	Senior Research Associate
ORCID	0009-0007-9751-2879
% FTE Commitment	20.00%

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APPLICATION DETAILS

Scientific Abstract
Continuous physiological monitoring is critical for managing babies in neonatal intensive care (NICU). However, conventional wired monitoring systems create a physical and psychological barrier preventing parent-infant physical contact during this critical developmental window. Evidence shows that this impacts on breastfeeding rates, length of stay, and neurodevelopmental outcomes and maternal mental health(CORE20plus5).
Innovation and Design
To address this challenge the Neonatal Wireless Transmission System (NeWTS) is a modular, wireless platform for monitoring vital signs: temperature, electrocardiogram (ECG), respiratory rate, and oxygen saturation. It functions as a wireless bridge between sensors and standard monitors. The proof-of-concept was tested in 34 NICU patients, and the project has secured funding for development to meet regulatory standards. By integrating with existing equipment and electronic medical records (EMRs), it supports NHS Net Zero goals (avoiding obsolescence). Improving quality of care whilst reducing costs. Our application to THRIVE aims to support us in our next step of clinical implementation.
We hope that the THRIVE programme will provide education and training to support commercialisation alongside the following work packages on clinical implementation.

WP1: Clinical Engagement with early adopters will gather real-world feedback from five diverse NICUs to understand local barriers to adoption.

WP2: Commercial Business Case (Health Innovation East)

Evidence Review and Gap Analysis will guide the necessary data collection to support the value proposition in the path for NICE approval. Business Case Development includes stakeholder workshop with NHS clinicians and commissioners to validate need, align with NICE guidance, and develop a scalable NHS business case. Social Impact and Sustainability focuses on compliance with the NHS Evergreen Sustainable Supplier Framework and integrates benchmarking against the Social Impact and Improvement Model (SIIM). Market Access and Procurement outlines pathways across local, regional, and national frameworks, ensuring readiness for various NHS purchasing models

WP3: Branding and Marketing will allow us to access marketing and creative expertise to create marketing materials for media communication.

Equity and Impact:
NeWTS specifically addresses the unmet needs of preterm infants, a population often excluded from medical innovations due to commercial perceptions of high-risk low reward. Babies and families from disadvantaged and minority backgrounds, are overrepresented in NICU, and will gain in terms of both physical and psychological health.

Dissemination and Stakeholder Engagement
Our co-production model includes a cross-sector advisory board comprising parents, clinicians, engineers, and industry experts. Dissemination will span academic channels and public engagement (supported by BLISS).

Project Management and Commercialisation
The core management team—(clinical, technical, and project management)—is supported by Cambridge Enterprise, guiding intellectual property protection. A patent has been filed and a freedom-to-operate analysis completed. The team would greatly value the opportunity in THRIVE to understand better the next steps in route to market.

Conclusion
The THRIVE programme would build expertise in a team committed to medtech development and addressing inequality of access to innovations in care for children.

Plain English Summary
The frustrations of a father, whose daughter spent her first 3 months in a Neonatal Intensive Care Unit highlighted the need for a wireless monitoring system. He found it difficult to interact with, and care for, her because of all the wires. Parents report that the wires create both a physical and emotional barrier to caring for their babies, and make nursing staff less keen to take babies out of incubators for them to hold. Importantly, physical contact improves brain development, reduces parental anxiety and stress and increases breast feeding rates.
Working with parents and staff within neonatal care, we have designed and tested a wireless prototype system for monitoring vital signs in neonatal intensive care (ECG, heart rate, temperature and oxygen levels) With support from NIHR, this academic prototype is being developed to ensure it meets the specifications and needs of a commercial product.
Removing the wires could reduce practical barriers to care for parents and save nursing time, enable parents to benefit from holding their babies, supporting breast feeding reduce length-of-stay.
We are on the path to manufacturing with IP filed, working on addressing the final technical challenges, and planning for clinical studies for regulatory submission. We have an established project advisory group including parents, clinical staff, and hospital management who are keen to ensure the project meets stakeholder's needs.
We would value the opportunity to take part in the entrepreneurial training, mentoring and networking to help us to identify the optimal route for commercialisation for NEWTS alongside undertaking work packages to provide data to increase commercial credibility. THRIVE would also help to build experience within the team to support other paediatric innovations.
WP1: Clinical Engagement: identification of early adopters will gather real-world feedback from five diverse NICUs to understand local barriers to adoption. This will provide opportunities for clinical feedback and engagement with hospital management and procurement teams with respect to purchasing and identifying barriers to adoption at the local level.
WP2: Commercial Business Case (Health Innovation East)
Evidence Review and Gap Analysis will guide the necessary data collection to support the value proposition in the path for NICE approval. Business Case Development includes stakeholder workshop with NHS clinicians and commissioners to validate need, align with NICE guidance, and develop a scalable NHS business case. Social Impact and Sustainability focuses on compliance with the NHS Evergreen Sustainable Supplier Framework and integrates benchmarking against the Social Impact and Improvement Model.

WP3: Branding and Marketing will allow us to access marketing and creative expertise.
There is a higher bar for derisking devices prior to leveraging commercial funding for technologies for paediatrics and intensive care as these areas are seen as high risk. This means babies, particularly those with the greatest needs are often excluded from the benefits of innovations in health care. Babies from disadvantaged and minority backgrounds, are overrepresented in NICU, and NEWTS could impact on parental mental health addresses one of the CORE20plus5 priorities of the NHS.

Changes from Previous Stage

Working with People and Communities Summary
This project is a direct result of parental feedback, and views of parents and other stakeholders have been central to the development. This continues through our stakeholder advisory board and partnership with our hospital maternity and neonatal voices group. KP is the project’s PPI lead. As a former NICU parent she has first-hand experience of the challenges faced by parents during this stressful period. In

her multiple roles as a neonatal service user representative, she also provides a key link to a diverse range of parents whose experiences and views will inform the successful development of NeWTS. The project follows a co-production model, directly involving parents throughout. KP has been actively involved in project management and strategy.

We have support from the UK's largest neonatal charity, Bliss. They provide a voice for parents, and also exert political leverage on expectations of quality in neonatal care.

PPI has been involved in all aspects of the innovation journey including.

System Design ensuring that the design meets the needs of parents

Clinical Study

The Clinical Study is being co-designed and delivered with PPI representation, refining its objectives (including ePROs), methodology, data collection tools, recruitment strategies considering diversity and accessibility, patient information materials, and parent-friendly summaries. The clinical studies will include diverse range of skin tones.

Business Development

To align the value proposition with service user needs, we are collaborating closely with PPI representatives and Bliss ensuring it addresses the unique concerns and requirements of users and the wider public. To broaden our inclusivity as part of the THRIVE application we wish to take the working prototype to a range of hospitals of different sizes and serving a diverse population. This will provide opportunities for parent and clinical feedback on system design and provide better understanding of training requirements and barriers to adoption at the local level.

We have a Project Advisory Board (PAB) who meet quarterly comprising:

This includes key stakeholders including former NICU parents, a senior neonatal nurse, neonatologist, Head of Clinical Engineering, Deputy Medical Director, medical devices engineer, and medical devices businessperson. Providing external oversight and advice

for the project around key decision-making.

SUMMARY BUDGET

Justification of Costs
We are requesting funding to support the following components critical to the delivery and implementation of the proposed project.

Clinical Lead (2 day/week for 9 months)
The Lead will provide strategic clinical oversight and leadership throughout the project by leading engagement with early adopter sites, supporting demonstrations and training for clinical teams. They will contribute to the development of business cases and implementation models, ensuring alignment with

NHS priorities, which is critical for adoption.

Research Staff (1 day/week for 9 months)
Research staff will support: collection and analysis of feedback from early adopter sites, evaluation of implementation processes and barriers, documentation of site visit outcomes to inform business case development.

Project Manager (1 day/week for 9 months)
The Project Manager will coordinate all aspects of the project, including logistics for training, site visits, and stakeholder engagement. They will ensure that outputs are aligned with THRIVE milestones, manage reporting, and facilitate effective communication between sites and partners.

Clinical Team (1 day/week for 9 months)
Members of the clinical team will engage in early adopter site visits, providing demonstrations of the system, training on use, and gathering insights into local procurement pathways. Their involvement is essential for co-developing implementation strategies tailored to NHS Trust needs.

Travel and subsistence costs are included to support face-to-face engagement with early adopter sites. These visits are essential for demonstrating the system, delivering in-person training, and building relationships with local champions. Budgeted items include return travel (rail or car) for staff, overnight accommodation where required due to geographic spread, subsistence in line with the guidelines.

Consumables

Consumables are requested to support administrative and operational project delivery. This includes materials for in-person events, printing of training and marketing materials, and technology licenses (if needed) for hosting training content or collaborative platforms.

Knowledge mobilisation, dissemination and impact

This item will support external marketing expertise that will be contracted to create a professional brand identity and develop communication tools, including a website, explainer videos, and downloadable guides. This will aid in stakeholder engagement and support adoption beyond the pilot sites. It will also support the development of online training modules such as short instructional videos and FAQs documents to support onboarding of new users at different sites.

Training and development – cost of the trainings, mentoring and travel delivered by NIHR.

Working with people and communities - A workshop will be organised involving patients, families, carers, and representatives from underserved communities to inform usability and inclusivity. This workshop will be co-produced and is critical for aligning the system with health equity aims. Costs include venue hire, facilitation, participant reimbursements, and catering.

HEI indirect costs for the partner University of Cambridge.

Other partner organization indirect costs -

We plan to work with Health Innovation East to develop business cases—both generic and Trust-specific—aligned with NHS procurement processes. This work is essential for translation into practice and long-term sustainability.

All the planned activities to be delivered by Health Innovation East are different from the activities funded by NIHR i4i grant (delivered by Cheata and Health Innovation East) which are focusing on value proposition and health economics. The planned activities under Thrive focus on creating a commercial

business case and entry to the NHS market.

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ACKNOWLEDGEMENT AND CONFLICTS

I have read and understood the terms of applying as Lead Applicant and accept this role	✓

Potential Conflicts	We currently hold an NIHR i4i PD award: NeWTS: Advancing Neonatal Care with Wireless Technology: From Prototype to Commercial Product. However none of the activities included in this application are included in that grant. Infact this application complements the product journey in supporting the team to gain further knowledge and insights in the route to translation into the clincal care pathway. As part of product development a patent for the core system functionality has been filed and is owned by Cambridge University Addenbrookes Hospital.

Lead Applicant:	Kathryn Beardsall	**Application Number:**	NIHR504679
Contracting Organisation:	CAMBRIDGE UNIVERSITY HOSPITALS NHS FOUNDATION TRUST		
Proposed Start Date:	01/11/2025	**End Date:**	31/07/2026

	Year 1	Total Cost to NIHR
STAFF COSTS		
Lead and co-applicants	£27,795.00	**£27,795.00**
Research staff	£25,823.00	**£25,823.00**
Other staff	£8,570.94	**£8,570.94**
Total staff costs	**£62,188.94**	**£62,188.94**
OTHER RESEARCH/ AWARD COSTS		
Travel and subsistence	£5,000.00	**£5,000.00**
Equipment	£0.00	**£0.00**
Consumables	£500.00	**£500.00**
Knowledge mobilisation, dissemination and impact	£6,000.00	**£6,000.00**
Training and development	£20,000.00	**£20,000.00**
Research inclusion	£0.00	**£0.00**
Working with people and communities	£2,000.00	**£2,000.00**
Other direct costs	£0.00	**£0.00**
HEI indirect costs	£15,273.90	**£15,273.90**
Other partner organisation indirect costs	£30,000.00	**£30,000.00**
TOTAL COST TO NIHR	**£140,962.84**	**£140,962.84**

Co-funding from other sources		**£0.00**
Cost to NIHR		**£140,962.84**
NHS support costs	£0.00	**£0.00**
NHS excess treatment costs	£0.00	**£0.00**
Non-NHS excess treatment costs	£0.00	**£0.00**
TOTAL FUNDING		

Application Budget: NIHR504679

REQUESTED	£140,962.84	**£140,962.84**

Neonatal Wireless Monitoring: Strategy for NHS Implementation

A father of a preterm baby in a neonatal unit (NICU) was frustrated that monitoring wires were preventing him from interacting with his child. Given the prevalence of wireless consumer devices, he questioned why similar technology was not available for his daughter.

Neonatal intensive care (NICU) has to combine the monitoring of vital signs whilst also addressing the developmental and psychological needs of babies and their parents. This is challenging for preterm infants who are extremely small and vulnerable, with fragile skin and who may be in hospital for many months. Parents and staff have reported that wired sensors create both physical and psychological barriers to care, and limit physical contact. Increasing physical contact has been shown to increase breastfeeding rates, reduce length of stay, and improve neurodevelopmental outcomes for babies. Therefore, the introduction of wireless monitoring could result in improved quality of care and savings for the NHS. This has been highlighted by the British Association of Perinatal Medicine in their submission to the NHS 10-year plan consultation and highlighting 'more research in this area is required to develop systems which are safe and effective'. https://www.bapm.org/articles/bapm-response-to-change-nhs

We have designed the Neonatal Wireless Transmission System, NeWTS, a novel and bespoke wireless bridge between vital sign sensors and the patient monitor. NeWTS includes modules for 4 vital signs: temperature, electrocardiogram (ECG and heart rate), respiratory rate, and oxygen saturation. Our proof-of-concept has been successfully tested with 34 babies in NICU, and we have NIHR i4i funding to realise this design to fulfil all user and regulatory requirements (CE/UKCA/FDA) for manufacturing. Clinical studies will be undertaken in 2026. Following submission for CE marking the path to implementation into the NHS is complex. ***The project team would greatly value the opportunity that THRIVE provides for a structured programme of entrepreneurial training, mentoring, peer support and networking to help us to identify the optimal route for commercialisation and build the case to leverage commercial support.***

There are increased challenges of moving from academic to commercial product in this area as technologies for both intensive care and the newborn are seen commercially as high risk relative to reward. This means babies, particularly those

with the greatest needs are often excluded from the benefits of innovations in health care and few devices are specifically designed for their needs. Off license use is common placing babies at risk. Furthermore, babies from disadvantaged families and ethnic minorities are also proportionately overrepresented in admissions to NICU. Optimising interventions at this time are, therefore, critical to reduce the risk of amplification of inequalities in health for these children. The impact on parental health addresses two of the CORE20plus5 priorities of the NHS (mental health and maternity).

Aims and objectives

NEWTS: To reduce the physical and psychological barriers between babies and their carers.

THRIVE programme: To utilise the resources in THRIVE to deliver a comprehensive business strategy incorporating i. engagement with local clinical teams ii. data to support a robust business plan and iii. Marketing strategy.
I fully commit to the training in person but also wish to use this training to build a team with expertise and enthusiasm to drive innovations in care. For timelines and milestones please see GANT chart.

Plan

As we move to realisation of a final product design for manufacturing, we have identified specific areas of work that would significantly derisk translation into clinical care and support the next steps in the commercial strategy. Furthermore, in supporting the NEWTS innovation the team would value the opportunity for this training to build wider knowledge within the team. This will not only guide the development of this innovation but build experience and expertise in a team committed to support further innovations in paediatric care.

WP1 Clinical Market and Early Adopters

Given the diversity of populations and health care delivery pathways across the NHS we wish to extend our planned usability testing to explore barriers to implementation. This will be important to fully understand the practicalities and barriers to adoption into clinical practice at the local level in the context of cultural and ethical differences. We will take the working prototype and associated documentation to a selection of 5 hospitals with different levels and sizes of NICU.

Deliverables will include reports from discussions with local clinical teams for feedback on IFUs and system design, hospital management and procurement teams, within a diverse range of hospital settings.

WP1 Commercial Business Case (Health Innovation East)

Evidence Review and Gap Analysis Having identified the key clinical and economic outcomes to focus on, through health economic modelling, Health Innovation East will work with a health economist to review existing evidence and perform a gap analysis. This will identify what evidence has been collected to date and what will be required to support successful entry into the NHS at scale. HIE will utilize years of regulatory experience to highlight the necessary evidence to collect in order to support the defined value proposition. This will support investment in collecting the right evidence, keeping costs down and timelines short.

Value Proposition and Development of NHS Business Case A business development workshop to enable the team to validate their innovation with experts from the NHS and understand the potential value proposition from NHS clinicians/commissioners' perspective. The workshop will have representation from NHS clinicians and commissioners across NHS Integrated Care Boards (ICB) and Health Innovation East. This provides an opportunity to communicate the unmet need and economic value to the NHS. To navigate and understand the implications of NICE endorsement, timelines and evidence required. To discuss and agree NHS policy alignment and enablers and understand stakeholder targets. The business development workshop would provide the background to better understand the market landscape, market opportunities, key policy barriers and levers, and the complexity of purchasing routes within the NHS national, regional, and local level. We will explore all of these as short- and long-term strategies.

This will support the population of an NHS business case template that can be adapted to support individual Trust and ICB applications for funding. Health Innovation East will put together a business case for a pilot in a local Integrated Care System. The pilot will aim to collect the required real-world evidence as the business case will pull together a clear vision for the use of NeWTS and how it aligns with NHS priorities. This will result in the development of a pre-filled business case template that can be adapted to support individual Trust and ICB applications for funding. It will include a literature review to demonstrate

alignment to NHS policy, validate the business case with an NHS Foundation Trust, funding options and health economics and cost and resource implications.

Social Impact Assessment Model and Sustainability. HIE is already committed to supporting with onboarding onto the NHS Evergreen Sustainable Supplier framework and developing the Carbon Reduction and Social Impact plans so that the company is well positioned for reporting on both direct and indirect emissions. In addition, it will be important to benchmark the NeWTS system and new legal entity policies, processes, against best practice and requirements of existing and upcoming legislation and policies against The Social Impact and Improvement Model (SIIM). The model covers themes across six category areas and will give a baseline in terms of maturity levels of sustainability within the organization. A detailed report containing target improvement areas, solutions, and recommendations. These will be prioritized and structured into an action plan to meet the supplier's needs and goals, while ensuring a proportionate response adhering with legislative requirements. This will give a clear overview of areas to prioritize and action.

Market access and procurement. A summary report on Framework opportunities will summarize the potential opportunities in terms of frameworks that are open for the product and provide helpful guidance on where to look for opportunities and what things to take into consideration such as Evergreen, Modern Slavery Assessment Tool, Carbon Reduction Planning

WP3 Branding and Marketing

This device will be used by clinical staff, parents and babies, it is essential that the device name and logo aligns with all three of these users. Gaining expertise in marketing to guide us on this will be essential to a successful product. Producing professional marketing resources for various media including marketing video will be critical.

Knowledge mobilisation, dissemination and impact The project management team and project advisory board are composed of patients, researchers, clinicians, NHS managers and industry experts in a co-production model. The system has been designed to support delivery of care in both tertiary and district general hospital in a range of settings. Whilst our Project Advisory Board formalises the input of key stakeholders, our network spans a wider range of expertise. This includes the local

and naitonal parent groups, clinicians from NICUs/Trusts across the country (of different size and specialism), Judge Business School, Cambridge Enterprise, HIE, the NHS Clinical Entrepreneur scheme, and UK Clinical Research Network (UKCRN). We will utilise direct public engagement and dissemination through social media, at key project milestones. Academic dissemination will include conference presentations and journal publications. We will promote article publications through the hospital press offices.

At a patient level, NeWTS improves quality of care, reducing barriers between babies and their parents. This will improve infants' physiological stability, reduce parental anxiety, and has the potential to increase breastfeeding rates and reduce length of stay. Impacts for staff include time saved in providing clinical care. These impacts are cost saving for the NHS as well as improving quality of care and addressing the drive to improve family-centred care. The impact on parental health addresses two of the CORE20plus5 priorities of the NHS (mental health and maternity).

Project management

The Office for Translational Research (UoC) provides translational project management to assist in meeting milestones within time and financial constraints, optimise partner interactions, facilitate regular project management meetings, risk reviews and assessments, updating risk registers, and preparing financial reports to the funder. The Project Management Team comprising Clinical lead, PPI lead, Technical lead and Project Management lead is PPI lead. The Project Advisory Board meet quarterly comprising: a former NICU parent, a senior neonatal nurse, neonatologist, Head of Clinical Engineering, Deputy Medical Director, medical devices engineer, and medical

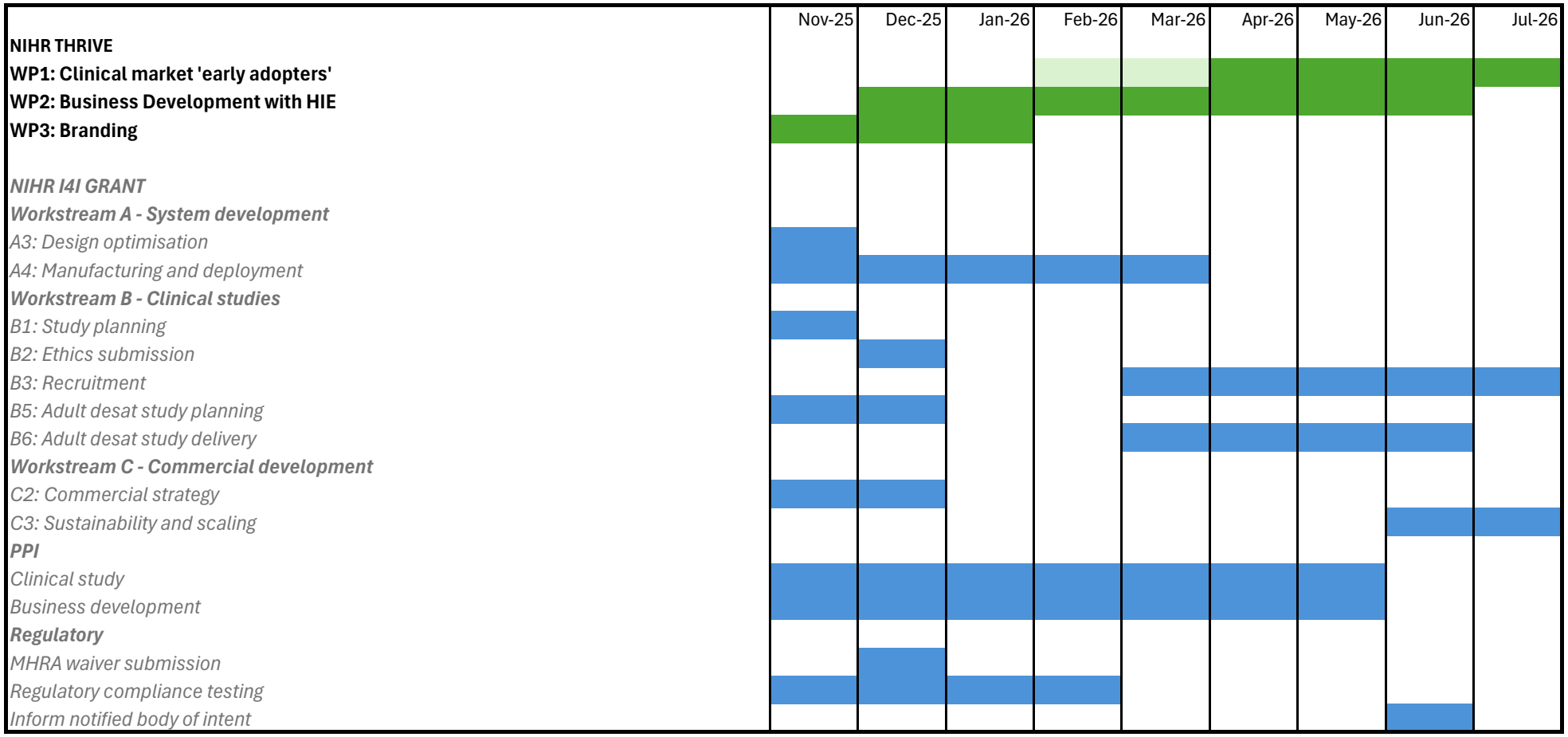
Intellectual property and commercialisation

Cambridge Enterprise (CE) is supporting the commercialisation strategy. CE are responsible for identifying, protecting, and managing intellectual property created during the project. A patent for the core system functionality has been filed. CE has conducted a freedom-to-operate (FTO) search and found that our system does not infringe on any existing patents. Our closest competitors are developing systems which are technologically different to NeWTS and do not present an FTO issue. CE will continue to monitor FTO. We have undertaken an extensive search of the

competitor landscape. We have a unique value proposition in that we can be integrated into existing monitoring systems that have already established links with EMR a key factor in the drive to digital health service and addressing the NHS Net Zero policy. Other systems do not demonstrate the battery life to allow integration into standard of care within the ward setting but are only suitable for short term use around delivery. Our system could work in both settings.

HIE will be able to support NeWTS to understand the procurement roadmap for the supply chain, and to ensure all areas are comprehensively addressed including policy levers, requirements, and how net zero/sustainability and healthcare improvement are intrinsically linked. We will leverage our existing relationships with clinicians to look at potential pilots for NeWTS. This will build case studies and evidence for future tenders as well as raising awareness with clinicians to make amendments to the current framework specifications. The regulatory landscape in the UK and Europe is in significant flux but we are working with regulatory experts GRS to develop a robust strategy that allows us to respond as the frameworks evolve. Our work to date including the proof-of-concept testing with babies in NICU. Wideblue, being part of Pivot International, is well-connected to supply chain experts, ensuring that component selection is sustainable to volume manufacture, and robust to fluctuations in supply.

We have not as yet determined the most appropriate commercialisation strategy but the resulting business will be committed to developing technology to improve the wellbeing of babies requiring medical care.


Nov-25
Dec-25
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Jul-26
NIHR THRIVE
WP1: Clinical market 'early adopters'
WP2: Business Development with HIE
WP3: Branding
NIHR I4I GRANT
Workstream A - System development
A3: Design optimisation
A4: Manufacturing and deployment
Workstream B - Clinical studies
B1: Study planning
B2: Ethics submission
B3: Recruitment
B5: Adult desat study planning
B6: Adult desat study delivery
Workstream C - Commercial development
C2: Commercial strategy
C3: Sustainability and scaling
PPI
Clinical study
Business development
Regulatory
MHRA waiver submission
Regulatory compliance testing
Inform notified body of intent

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Business Model Canvas NEWTS May 2025

Key Partners
- Wideblue
- Global Regulatory Services
- CHEATA
- Health Innovation Hubs
- Cambridge Enterprise
- BLISS (national preterm birth charity)
- University of Cambridge and Addenbrookes Hospital

Key Activities
- Clinical trials in NHS settings
- MHRA approval & compliance with UKCA marking
- ***Business Plan and gap analyses****
- ***Clinical Engagement****
- ***Branding****
- ***Establishment of Procurement path****
- Commercialisation strategy

Key Resources
- ***Clinical relationships with NHS Trusts****
- Compliance and regulatory experts (UKCA)
- Clinical Entrepreneur programme
- Skilled R&D and clinical teams
- Government innovation grants (e.g., NIHR)

Value Propositions
- Improved neonatal care
- Improved maternal mental health
- Increased breast feeding
- Reduced length of stay
- Addresses CORE20+5
- Reduces staff burden and increases care efficiency
- Supports NHS Long Term Plan (digital transformation)

Customer Relationships
- ***Engagement through NHS procurement programs****
- Clinical liaison officers assigned to Trusts
- Training sessions and CPD for clinicians
- Ongoing feedback for product refinement
- ***NICE technology evaluations & adoption pathways****

Channels
- NHS procurement portals
- ***Direct engagement with NHS Trusts****
- NHS Innovation Accelerator program
- Medical technology showcases
- ***Webinars and training through NHS Education****
- ***NIHR Thrive – Education and mentoring****

Customer Segments
- NHS Neonatal Intensive Care Units (NICUs)
- Specialist children's hospitals
- ***District general hospitals****
- Integrated Care Systems (ICSs)
- NHS innovation and transformation teams
- Global Market - US, India

Cost Structure
- Product development and certification (UKCA)
- Clinical evaluation in NHS settings
- Training and onboarding of NHS staff
- Integration with NHS systems
- Customer support and maintenance
- Ongoing data security compliance

Revenue Streams
- NHS direct procurement
- Framework contracts (e.g., NHS Supply Chain)
- Innovation grants and public funding
- NICE-supported adoption incentives
- Global market – US and India
- Private sector

*Activities specifically related to THRIVE